Weil Conjectures

Appendix D: Cohomology of torsion sheaves on curves

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D Cohomology of torsion sheaves on curves

The last step of proper base change for $\mathbf{P}^1$ compares the cohomology of a curve over two separably closed fields, one inside the other. Smooth base change needs the same comparison for affine curves, with coefficients prime to the characteristic. Both rest on Theorem 4.20*, whose assertions we recall. Let k be algebraically closed, let C be a separated scheme of finite type over k with $\dim C \leq 1$, and let F be a classical torsion abelian sheaf on $C_{\text{proét}}$. Then:

- (C_1) $H^q(C, F) = 0$ for $q > 2$;
- (C_2) if C is affine, $H^q(C, F) = 0$ for $q > 1$;
- (C_3) if $\operatorname{char} k = p > 0$ and F is p-primary, $H^q(C, F) = 0$ for $q > 1$;
- (C_4) if F is constructible with finite stalks and prime to $\operatorname{char} k$, all $H^q(C, F)$ are finite;
- (C_5) if C is proper and F is constructible with finite stalks, all $H^q(C, F)$ are finite;
- (C_6) if k'/k is an extension of algebraically closed fields and F is prime to $\operatorname{char} k$, the pullback maps $H^q(C, F) \to H^q(C_{k'}, F|)$ are isomorphisms;
- (C_7) if C is proper, the maps in (C_6) are isomorphisms for every torsion F;
- (C_8) for any open $U \subseteq C$, restriction $H^2(C, F) \to H^2(U, F|_U)$ is surjective.

Over $\mathbf{C}$ the vanishing and finiteness assertions are statements about a Riemann surface, which has the homotopy type of a CW complex of dimension two, or of one if it is not compact.

In algebraic geometry we compute the groups. Let C be a smooth proper connected curve of genus g and let n be invertible in k. Three ingredients compute the cohomology of μ_n: the

Kummer sequence $0 \to \mu_n \to \mathbf{G}_m \xrightarrow{n} \mathbf{G}_m \to 0$, the vanishing of $H^q(C, \mathbf{G}_m)$ for $q \geq 2$, which is a consequence of Tsen's theorem, and the Jacobian of Appendix C*. They give

$$H^0(C, \mu_n) = \mu_n(k), \qquad H^1(C, \mu_n) = J(C)[n](k) \simeq (\mathbf{Z}/n)^{2g}, \qquad H^2(C, \mu_n) = \mathbf{Z}/n$$

(Proposition D.2). These groups do not change under extension of algebraically closed fields, because torsion points of an abelian variety do not. For $\mathbf{P}^1$, which has $g = 0$, the computation fixes the normalization of the class of $\mathcal{O}(1)$ that Remark D.3 provides for the proof of proper base change. Removing a nonempty finite set of r points kills H^2 and raises the rank of H^1 to $2g + r - 1$ (Proposition D.4).

Proper base change, however, is needed for all torsion coefficients, including those divisible by the characteristic p, and there the answer changes. On the affine line, $H^1(\mathbf{A}^1_k, \mathbf{F}_p) = k[t]/\{h^p - h\}$ is infinite, and it grows when k is enlarged (Example D.7). Indeed, the Artin–Schreier torsors $y^p - y = at$, for $a \in k$, have pairwise distinct classes, and over a larger field k' there are new ones with $a \in k' \setminus k$. So for p-torsion coefficients, finiteness and invariance under field extension hold only for proper curves. This is why (C_5) and (C_7) of Theorem 4.20* assume properness, and why Lemma 4.21* is stated for proper curves.

The Artin–Schreier sequence $0 \to \mathbf{F}_p \to \mathbf{G}_a \xrightarrow{\varphi - 1} \mathbf{G}_a \to 0$ reduces the cohomology of $\mathbf{F}_p$ to the coherent cohomology of $\mathcal{O}_C$. On the affine line it presents H^1 as the cokernel of $h \mapsto h^p - h$ on $k[t]$, as above. On every curve it gives $H^q(C, \mathbf{F}_p) = 0$ for $q \geq 2$ (Theorem 4.20*, (C_3)).

A general torsion sheaf is reduced to a constant one by two observations. First, a sheaf supported on finitely many points has no cohomology in positive degrees. Hence passing to the normalization of a curve, or extending a sheaf by zero from a dense open subset, changes cohomology only in degrees zero and one, and only by finite groups (Lemmas D.10 and Lemma D.11). Second, a finite $\mathbf{F}_\ell$-local system becomes a successive extension of constant sheaves on a finite étale cover of degree d prime to ℓ. This cover is given by an ℓ-Sylow subgroup of the monodromy. Dividing the trace of the cover by d shows that the local system is a direct summand of the pushforward of its pullback (Lemma D.12). We follow the proof of [Stacks, 03SC].

Throughout this appendix, cohomology means cohomology on $C_{\text{pro}\acute{e}t}$, and the sheaves are classical in the sense of Definition 2.30*: by Theorem 2.32* such a sheaf is the inverse image $\nu_C^* G$ of a sheaf G on $C_{\acute{e}t}$ along the comparison morphism. By Theorem 2.35*, its cohomology, together with the restriction and pullback maps, agrees with étale cohomology, so we carry out all calculations on $C_{\acute{e}t}$ and transport them back by ν_C^*.

Let $p \geq 0$ be the characteristic of k. In this appendix ℓ is an arbitrary prime, possibly equal to p. In characteristic zero the hypotheses in (C_4) and (C_6) of Theorem 4.20* that F be prime to the characteristic hold automatically.

D.1 Kummer theory

Let C be a smooth curve over an algebraically closed field k, and let $\nu_C^* \mathbf{G}_m$ be the pullback of the multiplicative group $\mathbf{G}_m$ of $C_{\acute{e}t}$ to $C_{\text{pro}\acute{e}t}$. For n invertible in k, the Kummer sequence $0 \to \mu_n \to \mathbf{G}_m \xrightarrow{n} \mathbf{G}_m \to 0$ reduces the cohomology of μ_n to that of $\nu_C^* \mathbf{G}_m$, so we compute the latter first. In degree one, $H^1(C, \mathbf{G}_m)$ is the Picard group, as in the Zariski topology, while the groups $H^q(C, \mathbf{G}_m)$ vanish for $q \geq 2$ (Lemma D.1). In particular the Brauer group of C is zero. This matches, over $\mathbf{C}$, the topology of a Riemann surface, whose integral cohomology vanishes above degree two, and whose H^2 is accounted for by the degree of line bundles. In algebraic geometry the vanishing comes from Tsen's theorem, which says that forms of low degree over the function field $k(C)$ have nontrivial zeros. Over a finite field Tsen's theorem fails: by class field

theory the function field of a curve has nonzero Brauer group. This is one reason why we work over an algebraically closed field. With the lemma and the Jacobian, the Kummer sequence then computes the cohomology of μ_n on a proper curve (Proposition D.2).

Lemma D.1 (Cohomology of the multiplicative group)**.** *For a smooth separated curve C of finite type over an algebraically closed field, one has*

$$H^1(C, \nu_C^* \mathbf{G}_m) = \operatorname{Pic}(C), \qquad H^q(C, \nu_C^* \mathbf{G}_m) = 0 \quad (q \geq 2).$$

The identifications are natural under restriction and extension of the algebraically closed ground field. In particular $\operatorname{Br}(C) = H^2(C_{ét}, \mathbf{G}_m) = 0$.

Proof. We compute on $C_{ét}$. In degree one, descent for invertible modules identifies $\mathbf{G}_m$-torsors with line bundles. In degrees at least two, by contrast, the vanishing comes from Tsen's theorem, which we prove first (see also [Stacks, 03RM]). There are three steps: Tsen's theorem, proved by counting dimensions with Riemann–Roch; the vanishing of the Brauer group and of the Galois cohomology of $k(C)$; and the passage from the generic point to C by the divisor sequence.

Tsen's theorem says that $k(C)$ is C_1, that is, a homogeneous form of degree d in $m > d$ variables with coefficients in $k(C)$ has a nonzero zero. To see this, let $\overline{C}$ be the smooth proper model of $k(C)$, of genus g, choose an effective divisor D on $\overline{C}$ bounding the poles of all coefficients, and let the unknowns range over $H^0(\overline{C}, \mathcal{O}(NH))$ for an effective divisor H of positive degree. The value of the form then lies in $H^0(\overline{C}, \mathcal{O}(dNH + D))$. Once $N \deg H > 2g - 2$, we know both dimensions by Theorem C.3*. In terms of bases, the m unknowns amount to $m(N \deg H + 1 - g)$ unknowns in k. Each coordinate of the value, in a basis of the target space, is a homogeneous polynomial of degree d in these unknowns, so the vanishing of the value amounts to $dN \deg H + \deg D + 1 - g$ homogeneous polynomial equations in them. The number of unknowns minus the number of equations is $(m - d)N \deg H + (m - 1)(1 - g) - \deg D$, which is positive for large N because $m > d$. Since k is algebraically closed, homogeneous equations in more unknowns than equations have a nonzero solution [Stacks, 03RD].

The reduced norm form of a division algebra of degree $d > 1$ over $k(C)$ has degree d in $d^2 > d$ variables and does not vanish on a nonzero element. Therefore Tsen's theorem implies that $\operatorname{Br}(k(C)) = 0$ [Stacks, 03RF]. The same holds for every finite extension of $k(C)$, since it is again the function field of a curve over k. This is enough for Galois cohomology, because a field all of whose finite extensions have trivial Brauer group has $H^q(K, (K^{\mathrm{sep}})^\times) = 0$ for all $q \geq 1$ [Stacks, 03R8]; hence this holds for $K = k(C)$. To pass from $k(C)$ to C we use the divisor sequence, which relates $\mathbf{G}_m$ on $C_{ét}$ to the pushforward of the multiplicative group at the generic point and to skyscraper sheaves $\mathbf{Z}$ at the closed points. Of these, the former has no higher cohomology by this Galois vanishing, and the latter have none because the residue fields are algebraically closed. Therefore, from the long exact cohomology sequence, we conclude that $H^q(C_{ét}, \mathbf{G}_m) = 0$ for $q \geq 2$.

By Theorem 2.35* these identifications carry over to $C_{\mathrm{pro}ét}$, and they are natural because the comparison maps, the identification of torsors and the restriction maps are functorial. ∎

On a proper curve, the lemma and the Kummer sequence reduce the cohomology of μ_n to the groups $\operatorname{Pic}(C)[n]$ and $\operatorname{Pic}(C)/n$, both computed in Appendix C*.

Proposition D.2 (Smooth proper curves, torsion prime to the characteristic)**.** *Let C be smooth, proper and connected over an algebraically closed field k, with genus g, and let $n \geq 1$ be invertible in k. There are canonical identifications*

$$H^0(C,\mu_n) = \mu_n(k), \quad H^1(C,\mu_n) = J(C)[n](k),$$
$$H^2(C,\mu_n) = \mathbf{Z}/n, \quad H^q(C,\mu_n) = 0 \quad (q>2).$$

The isomorphism in degree two sends the Kummer boundary of a line bundle L to $\deg L \bmod n$. *These identifications commute with extension of algebraically closed fields.*

Proof. Pulling back the Kummer sequence of $C_{\acute{e}t}$ by ν_C^*, we get the exact sequence of sheaves on $C_{\mathrm{pro}\acute{e}t}$

$$0 \longrightarrow \mu_n \longrightarrow \nu_C^*\mathbf{G}_m \xrightarrow{(\cdot)^n} \nu_C^*\mathbf{G}_m \longrightarrow 0.$$

Here surjectivity of the nth power map is a local question. For a unit a the equation $T^n = a$ defines an étale cover, since its derivative nT^{n-1} is a unit. Moreover, the global units on the proper connected curve form the group $k^\times$, which is n-divisible because k is algebraically closed. Hence the long exact sequence starts with $H^0(C,\mu_n) = \mu_n(k)$. Since the map $H^0(C,\mathbf{G}_m) \to H^0(C,\mathbf{G}_m)$ is onto, $H^1(C,\mu_n)$ injects into $H^1(C,\mathbf{G}_m)$. Using Lemma D.1, we read off from the long exact sequence that

$$H^1(C,\mu_n) = \mathrm{Pic}(C)[n], \qquad H^2(C,\mu_n) = \mathrm{Pic}(C)/n,$$

with all higher groups zero.

Corollary C.12* gives the exact sequence

$$0 \longrightarrow J(C)(k) \longrightarrow \mathrm{Pic}(C) \xrightarrow{\deg} \mathbf{Z} \longrightarrow 0.$$

Multiplication by n on $J(C)(k)$ is surjective, with kernel $J(C)[n](k)$, and $J(C)[n]$ is finite étale of rank n^{2g} (Corollary C.15*). Applying the snake lemma to multiplication by n on this sequence, and using that $\mathbf{Z}$ is torsion free, we obtain $\mathrm{Pic}(C)[n] = J(C)[n](k)$ and $\mathrm{Pic}(C)/n = \mathbf{Z}/n$ via the degree. In particular, $H^1(C,\mu_n)$ is noncanonically isomorphic to $(\mathbf{Z}/n)^{2g}$ [Stacks, 03RQ].

Under an algebraically closed extension k'/k, the schemes $J(C)$ and $J(C)[n]$ base change to the corresponding objects for $C_{k'}$. The map $J(C)[n](k) \to J(C)[n](k')$ is bijective by Corollary C.15*. The same argument applies to the finite étale scheme μ_n. Finally, pullback of line bundles preserves the degree. Hence the pullback maps on cohomology are isomorphisms. ∎

Remark D.3 (Twists and the projective line)**.** The degree identifies $H^2(C,\mu_n)$ with $\mathbf{Z}/n$ without any choice. Since $\mu_n = (\mathbf{Z}/n)(1)$, twisting by -1 gives

$$H^2(C,\mathbf{Z}/n) = (\mathbf{Z}/n)(-1),$$

and identifying this group with $\mathbf{Z}/n$ requires the choice of a primitive nth root of unity. For $C = \mathbf{P}^1_k$ the Jacobian is zero, so the cohomology with constant coefficients is concentrated in degrees zero and two, and the Kummer boundary of $\mathcal{O}(1)$ corresponds to $1 \in \mathbf{Z}/n$. Section 4.4* uses this normalization on the geometric fibre.

D.2 Affine curves

Every smooth connected curve over an algebraically closed field that is not proper is affine and is the complement of a finite nonempty set of closed points in a smooth proper connected curve. To see this, take a projective model of its function field and normalize it, and recall that regular curves over a perfect field are smooth.

On the affine curve $U = \overline{C} \setminus D$ the Picard group is a quotient of $\mathrm{Pic}^0(\overline{C})$. It is therefore divisible, and $H^2(U, \mu_n) = 0$; this matches the fact that, topologically, a punctured Riemann surface is homotopy equivalent to a graph. In degree one, on the other hand, new torsors appear, since a Kummer torsor on U need not extend over $\overline{C}$. The obstruction to extending it at each puncture is the residue of the torsor, that is, the order modulo n of the corresponding rational section. For example, on $\mathbf{G}_m = \mathbf{P}^1 \setminus \{0, \infty\}$ the torsor $y^n = x$ has residues 1 and -1 at the two punctures. It generates $H^1(\mathbf{G}_m, \mu_n) \simeq \mathbf{Z}/n$. In general the residues sum to zero. The residue sequence of Proposition D.4 shows that $H^1(U, \mu_n)$ is free of rank $2g + \#D - 1$ over $\mathbf{Z}/n$. This is the first Betti number of a surface of genus g with $\#D$ punctures.

Proposition D.4 (Cohomology of a smooth affine curve)**.** *Let $\overline{C}$ be a smooth proper connected curve of genus g over an algebraically closed field k. Let $U = \overline{C} \setminus D$, where D is a finite nonempty set of closed points, and let n be invertible in k. There is a natural exact sequence*

$$0 \longrightarrow H^1(\overline{C}, \mu_n) \longrightarrow H^1(U, \mu_n) \xrightarrow{\partial} (\mathbf{Z}/n)^D \xrightarrow{\sum} \mathbf{Z}/n \longrightarrow 0. \tag{D.1}$$

Moreover, $H^0(U, \mu_n) = \mu_n(k)$ and $H^q(U, \mu_n) = 0$ for $q > 1$. The group $H^1(U, \mu_n)$ is non-canonically $(\mathbf{Z}/n)^{2g+\#D-1}$, and the pullback maps on all these groups are isomorphisms under algebraically closed extension.

Proof. By the Kummer sequence, the vanishing above degree one follows from the divisibility of $\mathrm{Pic}(U)$ and the vanishing of $H^q(U, \mathbf{G}_m)$ for $q \geq 2$ (Lemma D.1). For H^1 we describe μ_n-torsors as line bundles L with a trivialization of $L^{\otimes n}$. After extending L to $\overline{C}$, the orders at the punctures measure the failure of the torsor to extend. These orders give the residue map ∂. We prove exactness at each place, and then deduce the rank and the invariance under field extension.

Since $\overline{C}$ is regular, we may use divisors to see that

$$\mathrm{Pic}(U) = \mathrm{Pic}(\overline{C}) / \langle \mathcal{O}_{\overline{C}}(x) : x \in D \rangle.$$

One of these line bundles has degree one. Every class in $\mathrm{Pic}(\overline{C})$ is therefore congruent, modulo the subgroup generated by it, to a class of degree zero. Hence $J(\overline{C})(k)$ surjects onto $\mathrm{Pic}(U)$. As $J(\overline{C})(k)$ is n-divisible (Corollary C.15*), so is $\mathrm{Pic}(U)$. By the Kummer sequence and Lemma D.1, we have $H^2(U, \mu_n) = \mathrm{Pic}(U)/n = 0$ and $H^q(U, \mu_n) = 0$ for $q \geq 3$.

We construct the sequence (D.1) using line bundles. Recall that a μ_n-torsor on U is given by a pair $(L, \alpha : L^{\otimes n} \simeq \mathcal{O}_U)$ of a line bundle and a trivialization of its nth power. To define its residues, we extend L to a line bundle $\overline{L}$ on $\overline{C}$ and let $s = \alpha^{-1}(1)$ be the corresponding rational section of $\overline{L}^{\otimes n}$. The divisor of s is supported on D, since s generates on U. Its orders modulo n do not depend on the extension. Indeed, two extensions differ by $\mathcal{O}(E')$ with E' supported on D, and this changes the divisor of s by nE'. These orders define $\partial(L, \alpha)$. Their sum is zero modulo n, since the divisor of s has degree $\deg \overline{L}^{\otimes n} = n \deg \overline{L}$.

Suppose all orders are divisible by n, and write $\mathrm{div}(s) = nE$ with E supported on D. Then s is a nowhere vanishing section of $(\overline{L}(-E))^{\otimes n} = \overline{L}^{\otimes n}(-nE)$. So the trivialization extends to a torsor

with line bundle $\overline{L}(-E)$ on $\overline{C}$, and this torsor restricts to (L, α) on U. This proves exactness at $H^1(U, \mu_n)$.

To prove injectivity, note that a trivialization on U of a torsor on $\overline{C}$ gives a rational section of its line bundle whose nth power has no zero or pole on $\overline{C}$. Therefore the divisor of the section is zero, and the section trivializes the torsor on $\overline{C}$.

To see that ∂ maps onto the kernel of $\sum$, we lift a residue vector with sum zero modulo n to an integral divisor B supported on D. Its degree is divisible by n because the residues sum to zero. We claim that, as a consequence, there is a line bundle $\overline{L}$ with $\overline{L}^{\otimes n} \simeq \mathcal{O}_{\overline{C}}(B)$. To find it, choose $P \in \overline{C}(k)$ and put $L_1 = \mathcal{O}_{\overline{C}}((\deg B/n)P)$, so that the line bundle $\mathcal{O}_{\overline{C}}(B) \otimes L_1^{\otimes -n}$ has degree zero. By divisibility of $J(\overline{C})(k)$ it is $N^{\otimes n}$ for some N of degree zero, and $\overline{L} = L_1 \otimes N$ works. It remains to produce the torsor. The canonical section of $\mathcal{O}_{\overline{C}}(B)$ has divisor B and is nowhere zero on U. It therefore defines a trivialization of $\overline{L}^{\otimes n}|_U$, that is, a Kummer torsor on U whose residues are the orders of B. Finally, the summation map is surjective since D is nonempty. This proves that the sequence is exact (see also [Stacks, 03RR]).

The left term and the kernel of summation are free $\mathbf{Z}/n$-modules of ranks $2g$ and $\#D - 1$. Hence the resulting short exact sequence splits noncanonically. For naturality, note that the formation of the compactification, the boundary points, the line bundles and the orders of vanishing commutes with base change. The leftmost pullback map is an isomorphism by Proposition D.2, and the residue groups and their summation maps are unchanged. By the five lemma applied to the two exact sequences, the middle pullback map is an isomorphism. ∎

For the constant sheaf $\mathbf{F}_\ell$, with ℓ a prime different from $\operatorname{char} k$, the two preceding propositions prove (C_1)–(C_8) on smooth curves. Indeed, a primitive ℓth root of unity in k identifies $\mathbf{F}_\ell$ with μ_ℓ, compatibly with the pullback maps to $C_{k'}$, since the root of unity stays primitive in k'. For (C_8), note that an open in a smooth connected proper curve is either the whole curve or an affine curve, which has $H^2 = 0$, and that an open in a smooth affine curve is affine.

D.3 The Artin–Schreier sequence

Since the Kummer cover $T^n = a$ is étale only when n is invertible, p-torsion in characteristic p needs a different sequence. In the Artin–Schreier sequence the role of $\mathbf{G}_m$ is played by the additive group, whose étale cohomology is the coherent cohomology of the structure sheaf. Since coherent cohomology of a curve vanishes above degree one, this gives $H^q(C, \mathbf{F}_p) = 0$ for $q \geq 3$. For $q = 2$ we also need that $\varphi - 1$ is surjective on $H^1(C, \mathcal{O}_C)$, which holds over an algebraically closed field. So $H^2(C, \mathbf{F}_p) = 0$ even on a proper curve, and there is no analogue of the class of a point in H^2.

On a proper curve, $H^1(C, \mathbf{F}_p)$ is the fixed space of the Frobenius-linear map φ on the g-dimensional space $H^1(C, \mathcal{O}_C)$. Hence it has dimension at most g (Proposition D.5). The bound can be strict. For example, for an elliptic curve, φ on the line $H^1(E, \mathcal{O}_E)$ is bijective when E is ordinary, and zero when E is supersingular, as is $y^2 = x^3 - x$ over $\overline{\mathbf{F}}_3$. Accordingly $H^1(E, \mathbf{F}_p)$ has dimension one or zero. On an affine curve, by contrast, the cokernel of $\varphi - 1$ on the ring of functions contributes to H^1, and this group is infinite (Example D.7).

Proposition D.5 (p-torsion on a smooth proper curve in characteristic p)**.** *Let C be a smooth proper connected curve of genus g over an algebraically closed field of characteristic $p > 0$, and let φ be the Frobenius-linear endomorphism of $H^1(C, \mathcal{O}_C)$ induced by $a \mapsto a^p$ on $\mathcal{O}_C$. Then*

$$H^0(C, \mathbf{F}_p) = \mathbf{F}_p, \qquad H^1(C, \mathbf{F}_p) = \ker(\varphi - 1 : H^1(C, \mathcal{O}_C) \to H^1(C, \mathcal{O}_C)),$$

and $H^q(C, \mathbf{F}_p) = 0$ for $q > 1$. The group $H^1(C, \mathbf{F}_p)$ is finite, of $\mathbf{F}_p$-dimension at most g, and its pullback map is an isomorphism under algebraically closed extension.

Proof. The Artin–Schreier sequence turns the question into one about $\varphi - 1$ on $H^0(C, \mathcal{O}_C) = k$ and on $H^1(C, \mathcal{O}_C)$. The key fact is that on a finite-dimensional space over an algebraically closed field, a Frobenius-linear map minus the identity is surjective with finite kernel.

Step 1: the Artin–Schreier sequence. Write also φ for the endomorphism $a \mapsto a^p$ of the sheaf $\mathbf{G}_a$ on $C_{\acute{e}t}$. The Artin–Schreier sequence

$$0 \longrightarrow \mathbf{F}_p \longrightarrow \mathbf{G}_a \xrightarrow{\varphi - 1} \mathbf{G}_a \longrightarrow 0$$

is exact, because surjectivity is a local question and $T^p - T - a$ has derivative -1. The étale cohomology of $\mathbf{G}_a$ is the coherent cohomology of $\mathcal{O}_C$. Indeed, quasi-coherent sheaves have the same cohomology on the étale and on the Zariski site, by faithfully flat descent on affine opens and the Čech resolution. This coherent cohomology vanishes in degrees above one, since C is a curve. Since C is proper, $H^1(C, \mathcal{O}_C)$ is a finite-dimensional k-vector space.

Step 2: linear algebra. Let φ be a Frobenius-linear endomorphism of a finite-dimensional k-vector space V; we show that $\varphi - 1$ is surjective and that its kernel is the set of k-points of a finite étale scheme of order p^f with $f \leq \dim_k V$. To prove this, we reduce along a finite filtration to the part where φ is nilpotent and the part where φ is bijective. On the nilpotent part, $\varphi - 1$ is an automorphism given by an additive polynomial, with inverse $-(1 + \varphi + \varphi^2 + \cdots)$, a finite sum; in particular its kernel there is zero. On a bijective part of dimension f, by contrast, we choose a basis, invert the matrix of φ, and write the equations of the fibres as

$$x_i^p - \sum_j b_{ij} x_j = c_i \qquad (1 \leq i \leq f),$$

where (b_{ij}) is invertible. The coordinate algebra is free of dimension p^f, with basis the monomials having each exponent less than p. Moreover, the Jacobian matrix of these equations is $-(b_{ij})$, because $p = 0$ in k, and it is invertible; so each fibre is finite étale of degree p^f and has p^f points over the algebraically closed field k [Stacks, 0A3L]. In particular every fibre is nonempty, so $\varphi - 1$ is surjective, and the kernel has p^f elements.

Step 3: conclusion. We apply Step 2 to $H^0(C, \mathcal{O}_C) = k$ and to $H^1(C, \mathcal{O}_C)$ in the long exact sequence associated to the Artin–Schreier sequence. Surjectivity of $\varphi - 1$ on H^0 gives $H^1(C, \mathbf{F}_p) = \ker(\varphi - 1)$ on $H^1(C, \mathcal{O}_C)$, with the bound on its dimension, and surjectivity on H^1 together with $H^2(C, \mathcal{O}_C) = 0$ gives $H^q(C, \mathbf{F}_p) = 0$ for $q > 1$. Finally, the pullback map under an extension k'/k is an isomorphism [Stacks, 0HAN, 0A3P]: flat base change for coherent cohomology identifies the vector space and its Frobenius-linear map with their scalar extensions, and the kernel is a finite étale scheme, whose k-points and k'-points agree. As always, Theorem 2.35* carries the statement over to $C_{\text{pro\'et}}$. ∎

On an affine curve the same sequence is even simpler to use, because coherent cohomology vanishes in positive degrees. On the other hand, H^1 is now the cokernel of $\varphi - 1$ on an infinite-dimensional space.

Lemma D.6 (p-torsion on an affine curve in characteristic p)**.** *If U is affine over an algebraically closed field k of characteristic p, then, for the classical constant sheaf $\mathbf{F}_p$ and $\varphi(a) = a^p$ on $\Gamma(U, \mathcal{O}_U)$, we have*

$$H^1(U, \mathbf{F}_p) = \Gamma(U, \mathcal{O}_U)/(\varphi - 1)\Gamma(U, \mathcal{O}_U), \qquad H^q(U, \mathbf{F}_p) = 0 \quad (q > 1).$$

Proof. Quasi-coherent cohomology on an affine scheme vanishes in positive degrees, and, as in the proof of the proposition, étale and Zariski cohomology of $\mathbf{G}_a$ agree. In the long exact sequence associated to the Artin–Schreier sequence on $U_{ét}$, the terms $H^q(U, \mathbf{G}_a)$ for $q \geq 1$ are therefore zero. What remains is the exact sequence $0 \to H^0(U, \mathbf{F}_p) \to \Gamma(U, \mathcal{O}_U) \xrightarrow{\varphi - 1} \Gamma(U, \mathcal{O}_U) \to H^1(U, \mathbf{F}_p) \to 0$, together with $H^q(U, \mathbf{F}_p) = 0$ for $q > 1$. By Theorem 2.35* these statements carry over to $U_{\text{proét}}$. ∎

For the affine line we can compute the cokernel explicitly, and we see that it is infinite and grows with the field, unlike in the proper case.

Example D.7 (The affine line in characteristic p)**.** For $U = \mathbf{A}^1_k$ in characteristic p we get

$$H^1(U, \mathbf{F}_p) = k[t]/\{h^p - h : h \in k[t]\}.$$

Every class has a unique representative with zero constant term and all positive exponents prime to p. Indeed, we can remove exponents divisible by p using pth roots in the perfect field, and we can remove the constant term since $a \mapsto a^p - a$ is surjective on k. For uniqueness, note that a nonconstant polynomial $h^p - h$ has degree divisible by p. In particular the quotient is infinite, because the classes of t^m for positive m prime to p are $\mathbf{F}_p$-linearly independent. For an algebraically closed extension $k \subsetneq k'$ and $a \in k' \setminus k$, the class of at is not in the image of the quotient over k. Indeed, the reduction of a polynomial in $k[t]$ uses only pth roots and roots of $x^p - x = c$ in k, so its reduced representative lies in $k[t]$, whereas at is already reduced. Thus finiteness and invariance under field extension in degree one fail for p-torsion coefficients on an affine curve.

For the constant sheaf $\mathbf{F}_p$ on smooth curves, we obtain (C_1), (C_3), (C_5) and (C_7) from the proposition on proper curves, and (C_1), (C_2) and (C_3) from the lemma on affine curves, while (C_8) holds because H^2 vanishes in both cases.

D.4 Normalization

So far we have computed with constant coefficients on smooth curves, whereas Theorem 4.20* concerns arbitrary curves of dimension at most one and arbitrary torsion sheaves. Two reductions close the gap: normalization and removing finitely many points. Both rest on Lemma D.9, which says that a sheaf supported on finitely many points has no higher cohomology. Hence passing between a curve, its normalization and a dense open subset changes cohomology only in degrees zero and one, and only by finite groups that do not change under field extension.

For example, let $C : y^2 = x^3 + x^2$ be the nodal cubic. Its normalization $\mathbf{A}^1 \to C$ has two points over the node. For $M = \mathbf{Z}/n$ the normalization sequence of Lemma D.10 has cokernel $M^2/\Delta M \simeq M$ at the node. If n is invertible in k, then $H^1(\mathbf{A}^1, \mathbf{Z}/n) = 0$, and the long exact sequence gives $H^1(C, \mathbf{Z}/n) \simeq \mathbf{Z}/n$. This is the analogue of the loop created by the node.

Lemma D.8 (Finite morphisms and cohomology)**.** *Let $h : Y \to C$ be a finite morphism of schemes. The direct image h_* of classical sheaves is exact, its formation commutes with arbitrary base change, and $H^q(C, h_*G) = H^q(Y, G)$ for every classical abelian sheaf G on $Y_{\text{proét}}$ and every q.*

Proof. On the étale sites, Lemma 4.27* shows that h_* is exact, that its stalks are finite products of stalks of G, and that its formation commutes with arbitrary base change. Since h_* is exact, the Leray spectral sequence gives $H^q(C, h_*G) = H^q(Y, G)$. The comparison of classical direct

images and Theorem 2.35* transport these statements to the pro-étale site; see [Stacks, 03QP, 0959]. ■

Since a sheaf supported on finitely many points is a direct image from those points, it has no higher cohomology. All the reductions below use this vanishing.

Lemma D.9 (Sheaves with finite support)**.** *Let F be a classical abelian sheaf supported on finitely many closed points of a scheme C of finite type over an algebraically closed field k. Then $H^q(C,F)=0$ for $q>0$, and $H^0(C,F)$ is the direct sum of the stalks of F at these points; the pullback map on H^0 is an isomorphism under algebraically closed extension of k.*

Proof. The sheaf F is h_* of a sheaf on a finite disjoint union of copies of $\operatorname{Spec} k$, where h is the inclusion of the points. On these points taking sections is exact because k is algebraically closed, and the sections are the product, here a finite direct sum, of the stalks, which do not change under extension of k. Finally, Lemma D.8 carries these assertions over to C. ■

The normalization of a curve is finite and is an isomorphism away from finitely many points, so the constant sheaf on C differs from the direct image of the constant sheaf on C^{nor} only by a sheaf with finite support.

Lemma D.10 (Normalization of a constant sheaf)**.** *Let C be a reduced curve of finite type over an algebraically closed field, and let $\pi : C^{\mathrm{nor}} \to C$ be its normalization. For a finite abelian group M, the natural map*

$$\underline{M}_C \longrightarrow \pi_* \underline{M}_{C^{\mathrm{nor}}}$$

is injective with cokernel supported on finitely many points. At a point with b geometric branches, the cokernel is $M^b/\Delta M$, where Δ is the diagonal. These cokernels and the resulting exact sequences commute with algebraically closed field extension.

Proof. The map is computed on stalks by the formula for stalks of finite direct images, and the cokernel has finite support because π is an isomorphism away from finitely many points. Compatibility with field extension holds because normalization commutes with algebraically closed extension.

Normalization is finite for reduced schemes of finite type over a field. On each one-dimensional component, its normal model is regular, hence smooth over the perfect field, while the isolated components of dimension zero are reduced points, that is, copies of the spectrum of the ground field. Since the normalization is an isomorphism over a dense open of each component, the locus where it is not an isomorphism is finite.

By the calculation of stalks in Lemma 4.27*, the displayed map at a geometric point is the diagonal $M \to M^b$, with $b \geq 1$, so its kernel is zero and its cokernel has the asserted form. Where π is an isomorphism we have $b = 1$ and the cokernel vanishes; hence the cokernel is supported on the finite locus found above.

After algebraically closed extension, the normalization remains smooth and is a finite birational normal model of the reduced base change, so it is the normalization of the base change. Base change for finite morphisms identifies the fibres and their sets of branches, and therefore the diagonal maps and their cokernels are obtained by extension of scalars. ■

The normalization sequence reduces constant coefficients on a reduced curve to the smooth case. By Lemma D.9, its cokernel term has finite sections, unchanged under field extension, and no positive cohomology. Moreover, by Lemma D.8, $H^q(C, \pi_* \underline{M}) = H^q(C^{\mathrm{nor}}, \underline{M})$. The long exact sequence therefore yields isomorphisms $H^q(C, \underline{M}) \simeq H^q(C^{\mathrm{nor}}, \underline{M})$ for $q \geq 2$, while in degrees zero and one it gives a natural exact sequence relating C and C^{nor}, whose remaining term is a finite group. From these sequences, vanishing and finiteness pass from C^{nor} to C, and by the five lemma so does invariance under field extension.

The normalization of an affine or proper curve is again affine or proper. Hence all assertions of Theorem 4.20* for constant $\mathbf{F}_\ell$ follow from the calculations on smooth curves. For (C_8), we apply this comparison to C and to the inverse image of an open subset. The commutative square of restriction maps then reduces surjectivity to the smooth components.

D.5 Extension by zero

A constructible sheaf on a curve is locally constant on a dense open subset. Removing the finitely many remaining points replaces the sheaf by its extension by zero from this open, and changes cohomology only in degrees zero and one. Let $j : U \hookrightarrow C$ be a dense open in a reduced curve, and let $i : Z \hookrightarrow C$ be its finite complementary closed subscheme. For a classical sheaf F we have the exact sequence

$$0 \longrightarrow j_! j^* F \longrightarrow F \longrightarrow i_* i^* F \longrightarrow 0$$

of an open and its closed complement. We check its exactness on geometric stalks on $C_{\acute{e}t}$, and then pull it back.

Lemma D.11 (Removing finitely many points)**.** *Let C be a reduced separated curve of finite type over an algebraically closed field k, let $j : U \hookrightarrow C$ be a dense open with finite complement $i : Z \hookrightarrow C$, let ℓ be a prime, and let F be a classical constructible sheaf with $\ell F = 0$. For each of the assertions (C_1)–(C_8) of Theorem 4.20*, under its hypotheses on C and on the prime ℓ, the assertion holds for F if and only if it holds for $j_! j^* F$.*

Proof. The third term of the exact sequence displayed above is supported on the finite set Z, so by Lemma D.9 it has finite sections and no higher cohomology. Comparing the long exact sequences of F and $j_! j^* F$ then shows that each assertion passes in both directions.

The sheaf $j_! j^* F$ is again killed by ℓ, so the hypotheses on the coefficient prime are the same for both sheaves. Since the last term of the exact sequence has no positive cohomology, the long exact sequence yields

$$\begin{aligned} 0 \to H^0(C, j_! j^* F) \to H^0(C, F) \to H^0(Z, i^* F) \to \\ H^1(C, j_! j^* F) \to H^1(C, F) \to 0, \end{aligned}$$

and isomorphisms in degrees at least two. Since the sequence is natural under field extension and its term supported on Z is unchanged, the two long exact sequences show that the pullback maps are isomorphisms for F in all degrees if and only if they are isomorphisms for $j_! j^* F$ in all degrees. For (C_8), we apply the degree-two identifications after restriction to every open subset of C. See also [Stacks, 0GJA]. ∎

When j is dense, Lemma D.11, together with the calculation for constant $\mathbf{F}_\ell$ on C via Lemma D.10, shows that $j_!\underline{\mathbf{F}_\ell}$ satisfies the assertions of Theorem 4.20*. An arbitrary open is dense in the reduced union of the components it meets, a closed subscheme of C, so we can apply the dense case there and pass back to C: direct image along a finite morphism is exact [Stacks, 0A3Q] and, by Lemma D.8, does not change cohomology.

D.6 Reduction to constant coefficients

We reduce a local system to constant sheaves by pulling it back along a cover of degree prime to ℓ, on which it becomes unipotent. To find such a cover, note that a local system of finite $\mathbf{F}_\ell$-vector spaces has finite monodromy: the fundamental group acts on a fibre through a finite quotient. The local system then becomes a successive extension of constant sheaves on the cover defined by an ℓ-Sylow subgroup of the monodromy group.

On the full cover of its monodromy the local system even becomes constant. But that cover can have degree divisible by ℓ, and then the local system need not be a direct summand of the pushforward of its pullback. For example, let $v : y^2 = x$ be the double cover of $\mathbf{G}_m$, in characteristic different from two, and let $\ell = 2$. The composite of the unit $\mathbf{F}_2 \to v_*\mathbf{F}_2$ and the trace is multiplication by 2, which is zero. In fact $\mathbf{F}_2$ is not a direct summand of $v_*\mathbf{F}_2$ at all. The monodromy representation of $v_*\mathbf{F}_2$ is the regular representation $\mathbf{F}_2[\mathbf{Z}/2]$, which is indecomposable.

For a cover of degree prime to ℓ, on the other hand, Lemma D.12 shows that the trace divided by the degree splits the unit. It therefore suffices to prove the vanishing, finiteness and invariance statements for the pullback. The pullback is unipotent, that is, built from constant sheaves by extensions.

Lemma D.12 (Reduction to a unipotent local system)**.** *Let ℓ be a prime, let U be a connected scheme, and let G be a finite classical local system of $\mathbf{F}_\ell$-vector spaces. There is a finite étale cover $v : V \to U$ of degree d prime to ℓ such that v^*G has a finite filtration with successive quotients the constant sheaf $\mathbf{F}_\ell$. Moreover, G is a direct summand of v_*v^*G.*

Proof. The filtration comes from the representation theory of ℓ-groups, whose only simple module over $\mathbf{F}_\ell$ is the trivial one, and the splitting from the trace, whose composite with the unit is multiplication by the degree.

By Proposition 2.42*, G is the pullback of a finite étale local system, which corresponds to a finite continuous representation of the fundamental group. Let H be the finite image of this representation, choose an ℓ-Sylow subgroup P, and take the cover associated to its inverse image in the fundamental group. Its degree is $[H : P]$, which is prime to ℓ, and the restricted action factors through P. Since the augmentation ideal of $\mathbf{F}_\ell[P]$ is nilpotent, the simple modules of the finite ℓ-group P are trivial of dimension one, and a composition series yields the required filtration [Stacks, 0A3R].

On a fibre, the unit $G \to v_*v^*G$ is the diagonal and the trace map $\mathrm{Tr} : v_*v^*G \to G$ adds the d entries. Their composite is therefore multiplication by d, which is invertible on an $\mathbf{F}_\ell$-sheaf, so $d^{-1}\,\mathrm{Tr}$ is a retraction. ∎

To apply this to $j_!G$ on a reduced curve C, we take for U a dense smooth open on which G is locally constant. Normalizing C in the finite extensions of function fields defined by V, we get a finite morphism $\bar{v} : Y \to C$ with V the inverse image of U, and an open immersion $j' : V \hookrightarrow Y$. From the cartesian square formed by these maps and the formula for stalks of direct images along finite morphisms, we find that

$$j_! v_* v^* G = \bar{v}_* j'_! v^* G,$$

which is also the exchange formula of [Stacks, 0GJ1]. If C is proper, then Y is proper, and if C is affine, then Y is affine.

The assertions travel from constant sheaves to $j'_! v^* G$ on Y, then to $j_! v_* v^* G$ on C, and from there to the direct summand $j_! G$. Since $j'_!$ is exact, the filtration of $v^* G$ gives a filtration of $j'_! v^* G$ with graded pieces $j'_! \mathbf{F}_\ell$. The assertions of Theorem 4.20* hold for these pieces by the calculation for constant sheaves extended by zero. As we show at the start of the next section, they pass along extensions, so they hold for $j'_! v^* G$. Since $\bar{v}_*$ is exact, they hold for $j_! v_* v^* G$. By the splitting given by the trace, they hold for its direct summand $j_! G$. For invariance under field extension we use that the unit and the trace commute with extension of the ground field. Then the pullback map for $j_! G$ is a retract of the pullback map for $j_! v_* v^* G$, and a retract of an isomorphism is an isomorphism [Stacks, 03SD].

If C is not equidimensional, we choose a dense open that is smooth and is the disjoint union of the regular loci of the one-dimensional components and the isolated points. On the isolated points taking sections is exact, each connected one-dimensional piece is treated as above, and finite direct sums preserve all the assertions.

D.7 Dévissage

The assertions of Theorem 4.20* pass along short exact sequences. To see this, let $0 \to F_1 \to F \to F_2 \to 0$ be exact, and suppose that the assertions are known for F_1 and F_2. Vanishing and finiteness follow from the long exact sequence, and invariance under field extension follows from the map between the two long exact sequences and the five lemma. For (C_8) an extra argument is needed. Let $b \in H^2(U, F)$, and lift its image in $H^2(U, F_2)$ to $c_2 \in H^2(C, F_2)$. The obstruction to lifting c_2 to $H^2(C, F)$ lies in $H^3(C, F_1)$, which vanishes by (C_1), so such a lift c exists. The difference $b - c|_U$ comes from $H^2(U, F_1)$, and by (C_8) for F_1 it lifts to $H^2(C, F_1)$; adding that lift to c gives the required global class [Stacks, 0A5C].

Proof of Theorem 4.20. For constructible sheaves we reduce to the explicit calculations, and arbitrary torsion sheaves are filtered colimits of these. The reduction has four steps: dévissage to constructible $\mathbf{F}_\ell$-sheaves, extension by zero from an open on which the sheaf is locally constant, a cover of degree prime to ℓ that makes it unipotent, and normalization, which leaves constant coefficients on smooth curves.

By Theorem 2.35*, we may work on $C_{\acute{e}t}$, and since a universal homeomorphism does not change the étale topos, we may replace C by C_{red}. A constructible torsion sheaf has a finite exponent, because it has finitely many isomorphism types of finite stalks. It is therefore the direct sum of its finitely many primary parts, and a finite filtration of an ℓ-primary part has constructible $\mathbf{F}_\ell$-sheaves as graded pieces. By the stability under extensions just proved, it suffices to treat such sheaves.

Choose a dense smooth open $j : U \hookrightarrow C$ on which the sheaf is finite locally constant. By Lemma D.11 we may replace the sheaf by its extension by zero. On the connected components of U, Lemma D.12 and finite direct images reduce us to constant $\mathbf{F}_\ell$ extended by zero on curves finite over C; this is allowed because the assertions pass to direct summands and because, by Lemma D.8, finite direct images are exact and do not change cohomology. Finally, normalization and Lemma D.9 reduce these sheaves to the calculations with constant coefficients on smooth curves, namely the Kummer sequence for $\ell \neq p$ and the Artin–Schreier sequence for $\ell = p$. Since all reductions preserve the hypothesis that C is proper or affine, the theorem holds for constructible torsion sheaves.

An arbitrary torsion sheaf on the qcqs scheme C is a filtered colimit of constructible torsion sheaves, and cohomology commutes with filtered colimits [Stacks, 03SA, 03Q5]. For torsion prime to the characteristic or p-primary torsion, we may take the constructible sheaves with the same kind of torsion, by splitting off primary parts before passing to the colimit. Pullback and restriction also commute with filtered colimits. Hence vanishing and the pullback isomorphisms pass to the colimit. So does surjectivity in (C_8), because a representative of a class already occurs at some stage. Finiteness is asserted only for constructible sheaves. Finally, Theorem 2.35* carries these conclusions and their maps over to $C_{\text{pro}\acute{e}t}$. ■

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References marked * are to parts of the blueprint this document leaves out:

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Theorem 2.32 Classical sheaves as the essential image, in I The six operations (`theorem-classical-sheaves-essential-image`)

Theorem 2.35 Bounded-below cohomology comparison, in I The six operations (`theorem-bounded-below-cohomology-comparison`)

Proposition 2.42 Locally constant sheaves of finite presentation, in I The six operations (`proposition-locally-constant-sheaves-finite-presentation`)

Section 4.4 The projective line, in I The six operations (`sec-The-projective-line`)

Theorem 4.20 Cohomology of curves, in I The six operations (`theorem-cohomology-curves`)

Lemma 4.21 Separably closed field extension, in I The six operations (`lemma-extension-separably-closed-ground-field`)

Lemma 4.27 Finite morphisms, in I The six operations (`lemma-finite-morphisms`)

Appendix C The Picard scheme of a curve, in Appendices (`chap-The-Picard-scheme-of-a-curve`)

Theorem C.3 Riemann–Roch and coherent Serre duality, in Appendices (`theorem-riemann-roch-coherent-serre-duality`)

Corollary C.12 Degree and the Jacobian, in Appendices (`corollary-degree-jacobian`)

Corollary C.15 Torsion of the Jacobian and field extension, in Appendices (`corollary-torsion-jacobian-field-extension`)

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